Empirical Geometry of Multi-Task Learning: L1-Discovered Complexity, Geodesic Transfer, and Flag Manifold Construction via Cumulative Curriculum

Eli Natti
enatti@gmail.com

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Abstract

We present experimental results that deploy the geometric-prior framework for neural-network architecture design in a trained multi-task system. Using a rotation-of-projection architecture (H_rotational) with $\exp(A)$-parameterized mixers on five structurally diverse MNIST-derived tasks, we report three findings, each a diagnostic read off the trained weights. First, measured against a designed prior, L1 regularization on the learned perturbation to a canonical rotation reads off which tasks the prior already solves: tasks that are linear combinations of token features (addition, spatial center-of-mass) drive the perturbation to zero at every layer. Such a zero must be *audited*: it can mean genuine triviality (spatial) or *proxy substitution* (addition reads a magnitude axis, not symbolic sums), and the finer per-layer counts carry real seed variance, so we report them as distributions rather than sharp integers. Second, transfer to a held-out task is predicted by representational *alignment* rather than rank: all backbones hold per-half identity rank $\sim$6, yet transfer accuracy varies from 0.45 to 0.82 with the orientation of that capacity relative to the transfer task's geodesic; among aligned backbones, flag *coherence* further modulates transfer. Third, cumulative curriculum training constructs a *measurable flag manifold*, a nested sequence of subspaces $V_1 \subset V_2 \subset \cdots \subset V_5$, with zero forgetting, positive backward transfer (classification $0.670 \to 0.891$), and an endpoint invariant to construction order; sequential replacement destroys the nesting and recovers catastrophic forgetting as its geometric signature. A mixer-variant comparison ties the results to the compact/non-compact split of $\exp(A)$. Together these establish measurable geometric quantities (meaning fraction, per-half alignment, and flag nesting) as diagnostics for multi-task representation quality, deployed here at interpretable scale; recovering established phenomena through this lens is validation, not weakness.

1 Introduction

The architecture-as-geometric-prior framework [Natti, 2026] proposes that neural network architectures do not learn manifolds but *are* manifold priors: specific patterns of constraints on three reversible computational primitives—permutation (routing), pointwise transform (local frame), and invertible mixer (interaction)—that determine which geometric structures are representable. Training fits parameters within these constraints; the designer's task is to choose constraints that match the data's true geometry.

Each primitive operates in two modes. In *intrinsic mode*, the primitive transforms, reorders, or couples state vectors within a fixed-dimensional space: the number and dimensionality of states are preserved. In *extrinsic mode*, the primitive changes the space itself: the permutation adds or removes states (pooling, upsampling), the pointwise transform embeds into higher-dimensional or projects to lower-dimensional spaces, and the mixer lifts interactions into multi-channel coupling (e.g., multi-head attention). Extrinsic operations are where architectures change representational dimensionality, and the Grassmannian $\mathrm{Gr}(k, D)$, the manifold of k-dimensional subspaces of $\mathbb{R}^D$, provides the natural parameterization for such changes: moving from k to $k+1$ effective dimensions corresponds to geodesic motion on the Grassmannian, with smooth rotation into a new orthogonal direction.

The experiments in this paper work at a *fixed* state dimension $d = 12$ and use the mixer's parameters to control motion on the Grassmannian $\mathrm{Gr}(k, d)$, that is, to change which k-dimensional subspace of $\mathbb{R}^d$ carries task-relevant information, without changing d itself. The data enters at $\mathrm{Gr}(3, 12)$: each state vector's three base features $(x, y, \text{intensity})$ occupy a 3-dimensional subspace of the 12-dimensional ambient space, with the remaining 9 dimensions initially empty. The mixer $M = \exp(\theta R_{\text{base}} + \varepsilon A_{\text{perturb}})$ operates on *pairs* of d-dimensional states (s_i, s_j) via a $2d \times 2d$ transformation whose block structure maps directly onto the intrinsic/extrinsic distinction. The *diagonal* blocks $(d \times d)$ perform *intrinsic* mixing: they transform each state within its currently occupied subspace, preserving the effective dimension k. The *off-diagonal* blocks perform *extrinsic* mixing: they couple information from s_j into previously unoccupied directions of s_i and vice versa, rotating the representation into new dimensions and increasing k. The rotation angle θ and perturbation magnitude ε jointly control the balance between these two modes, how much of the mixer's action preserves the existing subspace versus expanding it. The energy in the off-diagonal blocks, which we call the *Grassmannian occupancy*, measures the effective dimensionality k of the representation: a layer with high off-diagonal energy has rotated information into more of the available d directions, corresponding to motion from $\mathrm{Gr}(k, d)$ toward $\mathrm{Gr}(k', d)$ with $k' > k$. The *meaning fraction* μ_ℓ then measures what fraction of this occupied subspace is aligned with a specific task's output geodesic, how much of the k-dimensional content is actually useful for a given task.

This setup has a key advantage: it allows us to study genuine Grassmannian phenomena (subspace rotation, effective dimensional change, flag manifold construction) at a fixed, small ambient dimension ($d = 12$), with θ and ε as the control parameters for motion on $\mathrm{Gr}(k, d)$. The data's entry point at $\mathrm{Gr}(3, 12)$ provides 9 empty dimensions as workspace for the mixer stack to populate as needed. The L1 regularization on the perturbation then plays the role of a Grassmannian constraint discovery mechanism: it identifies which layers need to expand the occupied subspace (increase k via nonzero ε) and which are served by the canonical rotation alone (fixed k at the value set by θR_{base}).

With this setup, the framework makes several testable predictions about multi-task learning:

1. If the architectural prior (a canonical rotation R_{base}) is already correct for a task's geometry, then a learned perturbation should be unnecessary; regularization pressure should drive it to zero.

2. If multiple tasks share a backbone, each task's representation quality should be measurable via its *meaning fraction* μ_ℓ, the alignment between the backbone's off-diagonal energy and the task's output geodesic.

3. If tasks are added progressively to a shared backbone while maintaining all prior tasks in the loss, the process should construct a *flag manifold*, a nested sequence of subspaces $V_1 \subset V_2 \subset \cdots \subset V_n$. Removing prior tasks from the loss should violate the nesting and produce catastrophic forgetting.

4. Transfer to a held-out task should depend on *alignment* (orientation of the backbone's identity manifold relative to the new task's geodesic) rather than on pool *rank* (dimensionality of the

task head's output projection). Among aligned backbones, *coherence* of the flag structure should further predict transfer quality.

Paper 1 [Natti, 2026] established the theoretical framework with small-scale illustrative experiments: a unified mixer demonstrating that convolution and attention are constraint configurations of a single primitive, and Grassmannian geodesics showing that subspace rotation is lossless while projection is not. Those experiments demonstrated the *concepts*; they did not test the *predictions* above under controlled multi-task conditions.

This paper provides that test. We construct an architecture, H_rotational, that instantiates the primitive framework directly: a learned butterfly permutation (routing), a sparse-dense preconditioner (pointwise transform), and a rotation-of-projection mixer parameterized as $\exp(\theta R_{\text{base}} + \varepsilon A_{\text{perturb}})$ where $A_{\text{perturb}} = UV^T - VU^T$ is an antisymmetric perturbation subject to L1 regularization. We train this architecture under seven conditions (five single-task specialists, simultaneous multi-task, and cumulative curriculum) on five MNIST-derived tasks spanning a range of structural complexity: 10-class classification, two-digit addition, pairwise comparison, spatial center-of-mass regression, and parity detection.

The results confirm the four predictions and reveal structure the theory did not anticipate. Measured against the designed prior, L1 regularization reads off which tasks it already solves: addition and spatial drive the perturbation to *zero* at every layer (the canonical rotation alone suffices), and such a zero must be audited before it is read as "simple." The finer counts, how many layers a harder task recruits, carry real seed variance, so the reliable separations are the sufficiency reading and the total recruited capacity, not a sharp per-task integer.

The cumulative curriculum produces a particularly striking result. As tasks are added progressively (spatial → classification → addition → odd/even → comparison), two things happen simultaneously: (1) no previously-learned task degrades, and (2) earlier tasks *improve*, classification accuracy rises from 0.670 after its own training phase to 0.891 by the final phase. This is the geometric signature of flag manifold construction: each new task extends the representational subspace into new dimensions that are partially parallel to existing tasks' geodesics, reinforcing rather than competing with them. The final state converges to the same backbone geometry as simultaneous multi-task training, regardless of curriculum order.

We organize the paper as follows. Section 2 describes the architecture, tasks, and experimental conditions. Section 3 presents the L1-discovered complexity hierarchy. Section 4 analyzes transfer learning and the meaning fraction diagnostic. Section 5 develops the flag manifold interpretation of cumulative curriculum training. Section 6 compares mixer parameterization variants. Section 7 discusses implications for continual learning and automatic architecture discovery.

2 Experimental Setup

2.1 The H_rotational Architecture

The H_rotational architecture is a 5-layer network where each layer consists of four components applied sequentially:

1. **Sparse-Dense Preconditioner**: A learned diagonal scaling $A_1 \in \mathbb{R}^d$ followed by a dense $d \times d$ matrix A_2, normalizing input activations before mixing.

2. **Learned Butterfly Permutation**: A parametric permutation implemented as a product of sparse butterfly factors with learnable phase angles ϕ. This determines the routing pattern, which pairs of state dimensions interact at the subsequent mixer.

3. **Rotational Projection Mixer**: The core primitive, parameterized as

$$M = \exp\left(\theta_{\text{head}} \cdot R_{\text{base}} + \varepsilon \cdot (UV^T - VU^T)\right) \tag{1}$$

where $R_{\text{base}} = \begin{pmatrix} 0 & -I_d \\ I_d & 0 \end{pmatrix}$ is the canonical rotation (the architectural prior), $U, V \in \mathbb{R}^{d\times r}$ are rank-r factors ($r = 2$) defining the learned antisymmetric perturbation, ε is a scalar controlling perturbation magnitude, and θ_{head} is a per-head rotation angle.

4. **RMSNorm**: Root mean square normalization stabilizing activations between layers.

The architecture uses the *unified state vector* representation from Paper 1: position is treated as a feature, not as a separate indexing structure. Each pixel in a 20×20 center-cropped MNIST image becomes a state vector $s_i = (x_i, y_i, \text{intensity}_i) \in \mathbb{R}^3$, where (x_i, y_i) are normalized pixel coordinates and intensity_i is the grayscale value. A single image thus produces 400 state vectors; a pair of images (for two-digit tasks) is arranged on a 20×40 canvas producing 800 state vectors, each carrying its own positional and feature information.

The base feature dimension is 3, and these vectors are zero-padded into $\mathbb{R}^d$ with $d = 12$. The data therefore enters the network at $\text{Gr}(3, 12)$, a 3-dimensional occupied subspace within the 12-dimensional ambient space. The remaining 9 dimensions are initially empty; the mixer's task is to rotate information into these directions as needed, expanding the effective dimension k beyond 3 through the learned parameters θ and ε.

The mixer operates on pairs of d-dimensional state vectors, producing a $2d \times 2d$ invertible transformation. The canonical rotation R_{base} mixes paired tokens via a $\pi/2$ rotation, exposing in-phase $(s_i + s_j)/\sqrt{2}$ and quadrature $(s_j - s_i)/\sqrt{2}$ components. The perturbation $UV^T - VU^T$ is constrained to be antisymmetric, ensuring that M remains in a neighborhood of the orthogonal group and the transformation is approximately volume-preserving.

L1 regularization. We apply an L1 penalty (weight $\lambda = 0.1$) to the perturbation parameters U and V at every layer. This creates pressure toward $A_{\text{perturb}} = 0$, the canonical rotation alone, unless the task loss requires the perturbation to remain nonzero. The balance between task loss and L1 penalty automatically discovers the minimal perturbation complexity needed by each task.

Parameter count. The full backbone contains approximately 3,350 parameters: 2,000 in butterfly permutation phases (5×400), 720 in pointwise transforms ($5 \times 12 \times 12$), 480 in mixer U/V factors ($5 \times \text{rank-}2 \times 24$), 125 in rotation angles, and small contributions from preconditioner scaling and mixer ε values.

2.2 Task Suite

We construct five tasks from MNIST digit images, chosen to span a range of geometric complexity:

Task	Type	Output	Structural character
A: Classification	10-way categorical	$P(\text{digit})$	Fine-grained discrimination
B: Addition	Regression	$d_1 + d_2$	Arithmetic on magnitudes
C: Comparison	Binary	$d_1 > d_2$?	Relational judgment
D: Spatial	Regression	center-of-mass (x, y)	Linear function of positions
E: Odd/even	Binary	parity of digit	Global property of identity
F: Magnitude bucket (held out)	3-way categorical	small/medium/large	Coarse magnitude binning

Table 1: The six MNIST-derived tasks. Tasks A–E are used for training; Task F (magnitude bucket: digit $\in \{0-3\}$ = small, $\{4-6\}$ = medium, $\{7-9\}$ = large) is held out for transfer evaluation on frozen backbones.

All tasks share the same input representation (pairs of MNIST digits encoded as unified state vectors with base features $(x, y, \text{intensity})$, zero-padded into $\mathbb{R}^{12}$) and the same backbone architecture. Only the task-specific output heads differ. Single-digit tasks (classification, spatial) use the left half of the canvas (400 state vectors); two-digit tasks (addition, comparison, odd/even) use the full 800-vector canvas.

2.3 Training Conditions

Seven experimental conditions test different aspects of multi-task geometry:

- **A–E (single-task specialists)**: Each condition trains the full backbone on one task only. These establish ceiling performance and reveal per-task L1 patterns in isolation.
- **F_multi (simultaneous multi-task)**: All five tasks trained jointly from the start, with equal loss weighting. The backbone must serve all five output heads simultaneously.
- **G_seq (cumulative curriculum)**: Tasks added progressively in the order spatial → classification → addition → odd/even → comparison. Once a task is added, it remains in the active loss for all subsequent phases. A sequential variant (each phase *replaces* the previous task) serves as a catastrophic forgetting baseline.

All conditions use 3 random seeds (0, 1, 2). Training runs for 50 epochs with early exit at accuracy ≥ 0.90 for classification tasks. Transfer evaluation trains a new output head (Task 6: magnitude bucket, small/medium/large digit) for 10 epochs on a frozen backbone.

2.4 Mixer Variants

We compare four parameterizations of the mixer to isolate the contribution of the antisymmetric constraint:

- **skew_only**: $A = \theta R_{\text{base}} + UV^T - VU^T$ (pure antisymmetric perturbation, guaranteed rotational).
- **full_pade**: $A = \theta R_{\text{base}} + UV^T$ (general perturbation, including symmetric component). Exponentiated via Padé [1,1] (Cayley transform).
- **pade_symreg**: full_pade with symmetric regularization $\lambda_{\text{sym}} \|A_{\text{sym}}\|_F^2$ (weight 0.1) suppressing the symmetric eigenvalue component.

- **matexp_symreg**: pade_symreg with exact `torch.linalg.matrix_exp` replacing the Cayley approximation.

The primary results use skew_only for single-task and F_multi conditions, and pade_symreg for the cumulative curriculum (where it proved most effective). Section 6 compares all variants systematically.

2.5 Measurement Framework: Three Ranks and Alignment

We report three distinct rank measurements throughout this paper, each probing a different property of the learned representation. Conflating them leads to incorrect conclusions about backbone capacity and transferability; we define them carefully here.

Pool rank (operation rank). After the backbone produces 800 state vectors $\in \mathbb{R}^d$, the task head mean-pools them to a single vector $z \in \mathbb{R}^d$ and applies a linear projection to the output dimension. The effective rank of z across the test set is bounded by the task's output dimension: a scalar task (addition, comparison, odd/even) forces pool rank $= 1.0$; a 2D regression (spatial) yields pool rank $\approx 1.5-2.0$; a 10-class task yields pool rank $\approx 4-6$ (lower than 10 because the network compresses class structure into a continuous representation). Pool rank measures the *operation's* intrinsic dimensionality, not the backbone's capacity.

Per-half rank (identity rank). We pool the backbone output over a spatial subset, the left 400 state vectors (one digit's image region) or the right 400, and compute the effective rank of this per-half representation across the test set. This measures the backbone's capacity *within a single input region*, independent of the task head's projection. Empirically, all backbones in our experiments show per-half rank ≈ 6, regardless of whether the downstream task is scalar or categorical. Since each state vector carries its (x, y) position as a feature, the per-half split is computed by filtering state vectors by their positional coordinate ($x < 0.5$ for the left image region, $x \geq 0.5$ for the right).

Alignment. Two backbones with identical per-half rank can transfer very differently. We measure alignment via three quantities: (1) *per-half signal accuracy*, a logistic probe from per-half pooled features to the per-half digit class; (2) *leakage*, a cross-prediction probe (left features $\rightarrow$ right digit); and (3) *meaning fraction* μ_ℓ, the cosine-squared alignment between a layer's occupied subspace and a task head's column space. High per-half rank with low signal accuracy indicates the capacity exists but is oriented away from digit identity (e.g., toward spatial coordinates).

The measurement pipeline is:

$$\underbrace{\text{backbone}}_{\text{800 state vectors}} \xrightarrow{\text{spatial split}} \underbrace{\text{per-half pool}}_{\substack{\text{per-half rank} \\ \text{signal, leakage}}} \xrightarrow{\text{full pool}} \underbrace{\text{task head}}_{\text{pool rank}} \xrightarrow{\text{freeze + retrain}} \underbrace{\text{transfer head}}_{\text{transfer accuracy}} \tag{2}$$

Transfer accuracy depends on per-half rank and alignment, not pool rank. The transfer head pools the backbone's full output and fits a new linear projection, it reads whatever the backbone holds, not what the original task head projected. A backbone with pool rank 1 (scalar task) but per-half rank 6 (rich identity features) can transfer well if those identity features are aligned with the transfer task's geodesic.

3 L1 Sufficiency and the Substitution Audit

3.1 The Sufficiency Reading

The L1 analysis measures, relative to the designed prior, which tasks the canonical rotation already solves. The cleanest reading is at the low end: under L1 pressure the addition and spatial specialists drive the perturbation to *exactly* zero at every layer (layers 2–5 reach $\|U\| = \|V\| = 0.000$; layer 1 retains residual norms below 0.02, 85–90% sparse, contributing negligibly). The canonical rotation R_{base} alone, with no learned off-diagonal coupling, is sufficient for both. Sparsity as an unsupervised complexity signal is established [Frankle and Carbin, 2019, Louizos et al., 2018, Li et al., 2018, Aghajanyan et al., 2021]; what the primitive framing adds is that the count is measured *relative to a designed prior*, so a zero names a specific sufficient primitive stack rather than a generic parameter budget.

Above this zero floor, the surviving active-layer count carries real seed variance, so we report it over five seeds rather than as a single integer (Table 2). The counts trend the way the overparameterization picture predicts, the lone classification specialist folds the most layers, the multi-task conditions the fewest, but the ordering is not significant at five seeds (specialist vs. curriculum $\Delta = 1.2$ active layers, exact-permutation $p = 0.13$), so we report it as directional only. The one separation that reaches significance is in *total* recruited capacity, which distinguishes the specialist from the multi-task backbones (4.17 vs. 5.72, $p = 0.016$); seed profiles are rank-ordered before averaging, and the count is threshold-robust (L1 drives the per-layer norms bimodal, so sweeping the active threshold over 0.005–0.5 moves a single entry).

A preliminary single-seed run of this experiment reported sharp integer counts (addition/spatial 0/5, comparison 1/5, odd/even 4/5, classification 5/5, multi-task 4/5 with "one interior layer pure"). The five-seed rerun does not support the high-complexity end: the classification specialist's active-layer count is 3.6 ± 1.1 (range 2–5), so "5/5" reproduces in one of five seeds, and the specific layer that folds is seed noise. The reproducible content is the zero floor (addition, spatial), the seed-variable counts reported as distributions, and the significant total-capacity separation. Comparison and odd/even sit between the extremes but were not re-run at five seeds, so we do not assign them sharp counts; classification and odd/even remain the tasks that recruit the most perturbation, but the exact figure is not a stable integer.

Condition (5 seeds)	**Active /5 by seed**	**mean ± sd**	**Total cap.**
Classification specialist	$5, 3, 4, 4, 2$	3.6 ± 1.1	4.17
Simultaneous multi-task	$4, 5, 4, 4, 5$	4.4 ± 0.6	5.42
Cumulative curriculum	$4, 5, 5, 5, 5$	4.8 ± 0.5	5.72
Addition / spatial specialist	$0, 0, 0, 0, 0$	0.0	,

Table 2: Active perturbation layers over five seeds (`skew_only` mixer; layer active iff $\|U\|_F > 0.05$ or $\|V\|_F > 0.05$ after training). Addition and spatial hold 0/5 across seeds (robust). Counts vary within condition; the specialist-folds-more ordering is directional, not significant ($p = 0.13$), while total recruited capacity separates the specialist from the multi-task backbones ($p = 0.016$).

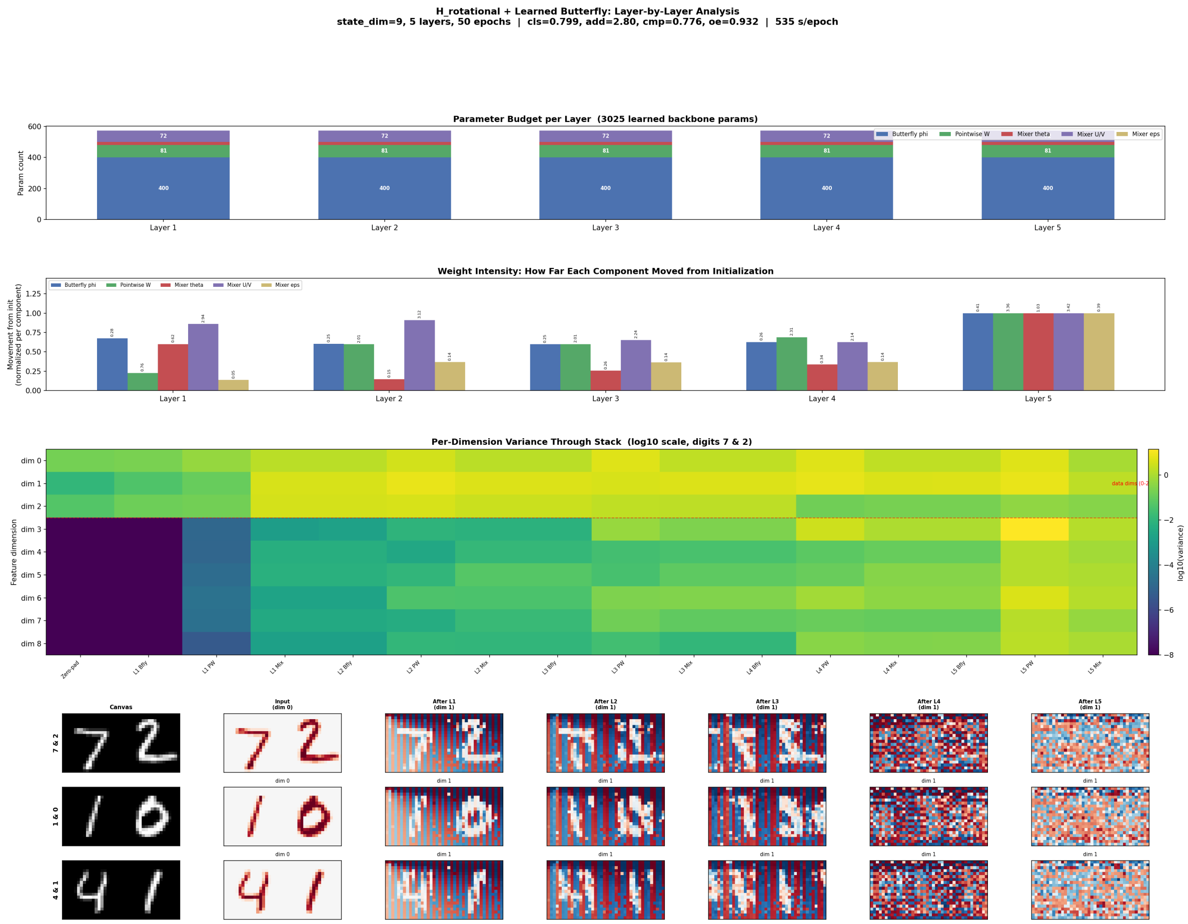

Figure 1: Per-layer perturbation activity across conditions. Addition and spatial are flat at zero at every layer (robust across seeds); harder tasks recruit perturbation with real seed-to-seed variation in how many layers, and which. The per-layer ε profile of any single run is not reproduced across seeds, so we read the counts as distributions (Table 2), not as a fixed per-task integer.

3.2 Why the Canonical Rotation Suffices for Linear Tasks

The zero-perturbation result for addition and spatial has a precise geometric explanation. The canonical rotation $R_{\text{base}} = \begin{pmatrix} 0 & -I_d \\ I_d & 0 \end{pmatrix}$ applied to a pair of state vectors (s_i, s_j) produces:

$$s'_i = -s_j \tag{3}$$
$$s'_j = s_i \tag{4}$$

In the rotated basis, the two natural linear combinations accessible without further transformation are:

$$a_{\text{sum}} = (s_i + s_j)/\sqrt{2} \quad \text{(in-phase component)} \tag{5}$$
$$a_{\text{diff}} = (s_j - s_i)/\sqrt{2} \quad \text{(quadrature component)} \tag{6}$$

If the state vectors embed a magnitude component (the scalar value of the digit, captured from pixel intensity or spatial extent), then the sum of two digits is directly accessible as a linear readout of a_{sum}. Similarly, center-of-mass is a linear function of positional features, accessible via the same in-phase component. No learned off-diagonal coupling is needed because the architectural prior (R_{base}) already exposes the task-relevant linear combination.

This is the cleanest possible validation of the architecture-as-geometric-prior principle. The same architecture, with the same initial capacity at every layer, is presented with five tasks. For two of them, the loss + L1 penalty jointly discover that the prior needs *zero correction*, the architecture's built-in rotation is exactly the transformation the task requires.

3.3 The Substitution Audit: Genuine Triviality vs. Proxy

The same 0/5 count carries different meanings for spatial and addition, and auditing which is where the diagnostic earns its keep. For spatial the solved and the requested task coincide: center-of-mass is a linear function of positions, computed directly by the canonical rotation. For addition they do not. The addition specialist reaches MAE ≈ 1.78 at pool rank 1.0 using only the canonical rotation, but it is not doing symbolic arithmetic: L1 pressure has substituted the nearest task the prior affords, a continuous magnitude readout, for digit-identification-then-sum. The evidence (error roughly uniform across digit pairs, a global-pool probe recovering only 3–4 magnitude bins, and near-unity CKA to the spatial backbone) is developed below; the audit is cheap, uses only trained weights, and should accompany any L1 zero before it is read as "simple."

The pool rank of 1.0 does *not* indicate a collapsed backbone. It reflects the operation's intrinsic dimensionality: addition is a scalar task, so the pool + linear head projects all backbone content onto one direction. Per-half probing (Section 2.5) reveals that the backbone holds a rank$\sim$6 identity manifold within each spatial region, comparable to what a classification specialist maintains, but the mean-pool operation projects this rich representation onto the single axis that linearly correlates with digit magnitude.

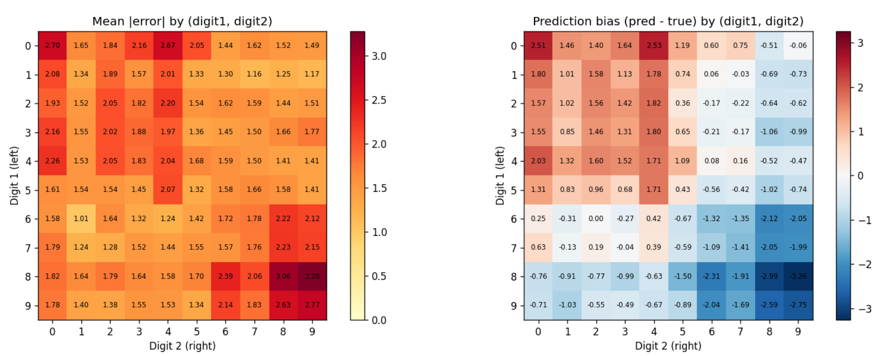

Figure 2: Addition error heatmap by digit pair (left digit $\times$ right digit). Errors are roughly uniform across pairs, consistent with a continuous magnitude readout from a richer identity manifold.

The backbone's internal richness is further revealed by how input geometry changes alignment without changing rank. Under the standard canvas encoding (two digits on a 20×40 canvas), per-half digit probe accuracy is $\sim$0.27 with substantial cross-region leakage ($\sim$0.42): the backbone holds identity features but the canvas geometry allows left and right features to mix freely. Under a paired encoding (two separate 20×20 images with an explicit side channel), per-half rank remains $\sim$6 but digit probe accuracy roughly doubles to $\sim$0.52 and leakage drops $\sim$4$\times$ to $\sim$0.12. Same rank, different alignment: the explicit side channel reorganizes the identity manifold so left and right digits occupy more disjoint subspaces.

Encoding	**Per-half rank**	**Digit probe acc**	**Leakage**	**Pool rank**
Canvas (20×40)	6.5	0.27	0.42	1.00
Paired (2×20×20 + side)	5.9	0.52	0.12	1.00

Table 3: B_add under two input encodings (200 epochs, seed 0). Pool rank is invariant (set by the scalar output). Per-half rank is similar (∼6). What changes is alignment: how the identity manifold is oriented relative to digit class and spatial region.

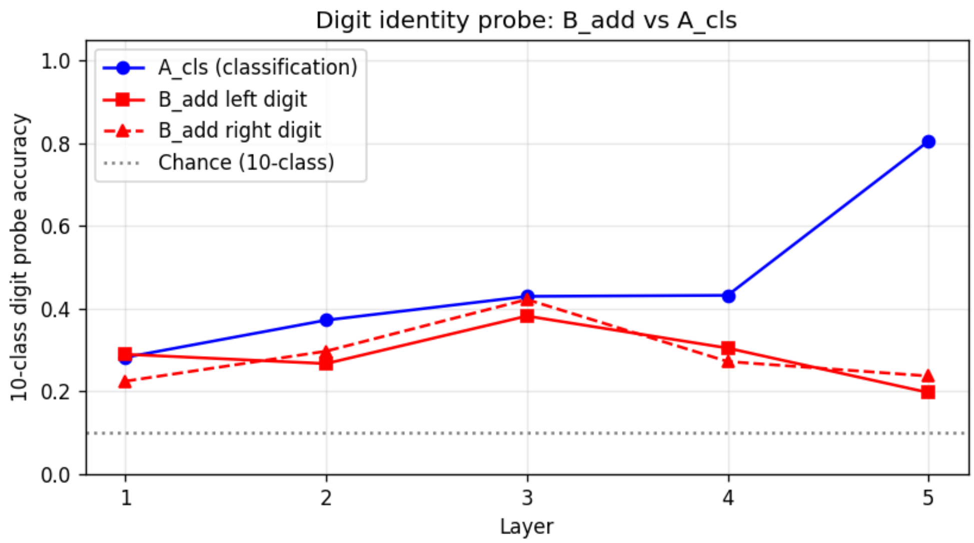

Figure 3: Linear probe accuracy for digit identity at each layer under canvas encoding. The addition specialist (B_add) achieves ∼35% probe accuracy on the global pool (3–4 effective magnitude bins), far below the classification specialist's ∼90%. However, per-half probing reveals rank∼6 identity features hidden behind the scalar pool projection.

CKA similarity between the addition and spatial backbones (Figure 4) shows near-identical layer-by-layer representations, confirming that both zero-perturbation tasks exploit the same geometric feature, a magnitude axis readable via canonical rotation. The per-half identity manifold is a byproduct of the backbone's representational capacity, not a feature the task explicitly requires.

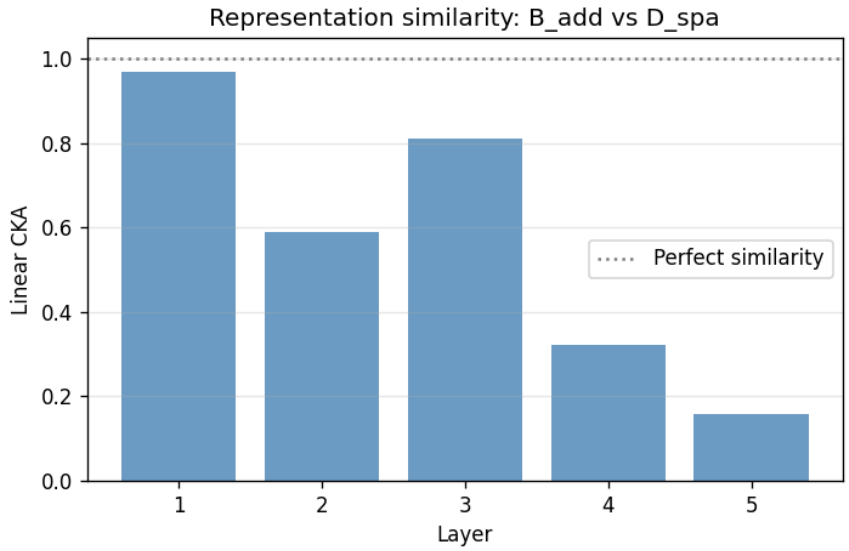

Figure 4: CKA similarity between addition (B_add) and spatial (D_spa) backbone representations at each layer. Near-unity similarity confirms both tasks learned the same 1D magnitude feature through the same canonical rotation pathway.

3.3.1 Addition vs. Subtraction: Algebraic Symmetry Shapes the Geodesic

Both addition ($a + b$) and subtraction ($a - b$) are scalar operations on pairs of digits, and both yield pool rank 1.0. Yet the two tasks produce measurably different geometric signatures in the backbone, revealing that the task's algebraic symmetry, not just its output dimensionality, shapes the mixer's utilization pattern.

Task	Per-half rank	Mixer eig rank profile	Transfer acc	MAE
B_add (paired)	5.8–6.1	[2.6, 4.1, **24.0**, 3.9, 6.7]	0.748	1.56
B_sub (paired)	3.5–5.0	[4.3, 3.4, 7.8, 4.5, 4.3]	0.573	,

Table 4: Addition vs. subtraction under paired encoding (200 epochs, seed 0). Both are scalar pool-rank-1 tasks. Addition concentrates mixer rotation at layer 3 (rank 24 spike); subtraction spreads it across all layers. Per-half rank and transfer accuracy are both lower for subtraction.

Addition is symmetric ($a + b = b + a$): the in-phase component $(s_i + s_j)/\sqrt{2}$ exposed by R_{base} directly encodes the answer. The mixer concentrates its rotation at a single layer (the rank-24 spike at layer 3), with the remaining layers near-idle. Subtraction is antisymmetric ($a - b = -(b - a)$): the answer requires the quadrature component $(s_j - s_i)/\sqrt{2}$, which must preserve the sign distinction between left and right. The mixer spreads its rotation across all five layers rather than concentrating it, and the per-half identity manifold is lower-rank (3.5–5.0 vs. 5.8–6.1).

The transfer consequence is direct: addition's per-half manifold encodes digit identity in a way that aligns with magnitude bucketing (the transfer task), yielding 0.748 accuracy. Subtraction's antisymmetric encoding compresses magnitude information across the sign boundary, yielding only 0.573. Same architecture, same input encoding, same pool rank, but the algebraic symmetry of the task reshapes the geodesic path through the Grassmannian and produces a different backbone geometry.

3.4 Multi-Task L1 Pattern

The multi-task backbone (F_multi) recruits perturbation at 4.4 ± 0.6 of five layers over five seeds, more than either zero-floor specialist, and directionally near the classification specialist rather than below it (Table 2). We do not read a clean "one interior layer pure" structure into it: which layer, if any, folds varies by seed, and the earlier single-seed observation of a pure interior layer with a maximal final layer is not reproduced across seeds. What *is* reproducible is that the multi-task backbones recruit more *total* capacity than the specialist (5.42 vs. 4.17, $p = 0.016$), consistent with a shared backbone needing more off-diagonal content to serve five heads at once.

The mixer parameterization determines whether serving five heads costs classification accuracy. Under `pade_symreg`, multi-task classification (0.880 ± 0.016) matches the classification specialist (0.889 ± 0.021) within noise at a comparable budget; under `skew_only`, both the specialist (0.816) and multi-task (0.752) classification are $\sim$7% worse, and the gap between them widens. This indicates that the `skew_only` constraint restricts the reachable geodesics, classification's optimal path uses symmetric mixer components, so the multi-task deficit under `skew_only` is a constraint problem, not an interference problem (Section 6). We do *not* claim the multi-task backbone matches the specialist "using one fewer perturbation layer": at five seeds the specialist's layer count is not reliably higher, and total recruited capacity in fact runs higher for the multi-task conditions.

4 Transfer and the Perpendicular Reservoir

4.1 Transfer to a Held-Out Task

To measure the generality of each backbone's learned representation, we freeze the backbone after training and train a new output head for a held-out sixth task: magnitude bucket (classifying digits as small: 0–3, medium: 4–6, or large: 7–9). This task was not seen during backbone training. We train for 10 epochs on the frozen backbone and report transfer accuracy.

Condition	Transfer accuracy	Pool rank	Per-half rank
A: Classification (specialist)	$\mathbf{0.815 \pm 0.018}$	4.09 ± 0.04	$\sim$6
F: Multi-task (joint)	0.811 ± 0.010	4.74 ± 0.35	$\sim$6
G_seq (cumulative, pade_symreg)	0.761 ± 0.053	4.81 ± 0.18	$\sim$6
G_seq (sequential)	0.681 ± 0.010	$\mathbf{5.45 \pm 0.32}$	$\sim$6
C: Comparison	0.686 ± 0.001	1.00 ± 0.00	$\sim$6
B: Addition	0.679 ± 0.022	1.00 ± 0.00	$\sim$6
D: Spatial	0.450 ± 0.025	1.48 ± 0.19	$\sim 6^{\dagger}$
E: Odd/even	0.443 ± 0.030	1.00 ± 0.00	$\sim$6

Table 5: Transfer accuracy, pool rank (bounded by output dimension), and per-half identity rank for each training condition. Per-half rank is $\sim$6 across all conditions, the backbone holds digit identity features regardless of the training task. Transfer depends on how those features are *aligned* with the transfer task, not on pool rank. †D_spa per-half rank is expected $\sim$6 based on architectural capacity; its features are oriented toward spatial coordinates rather than digit identity.

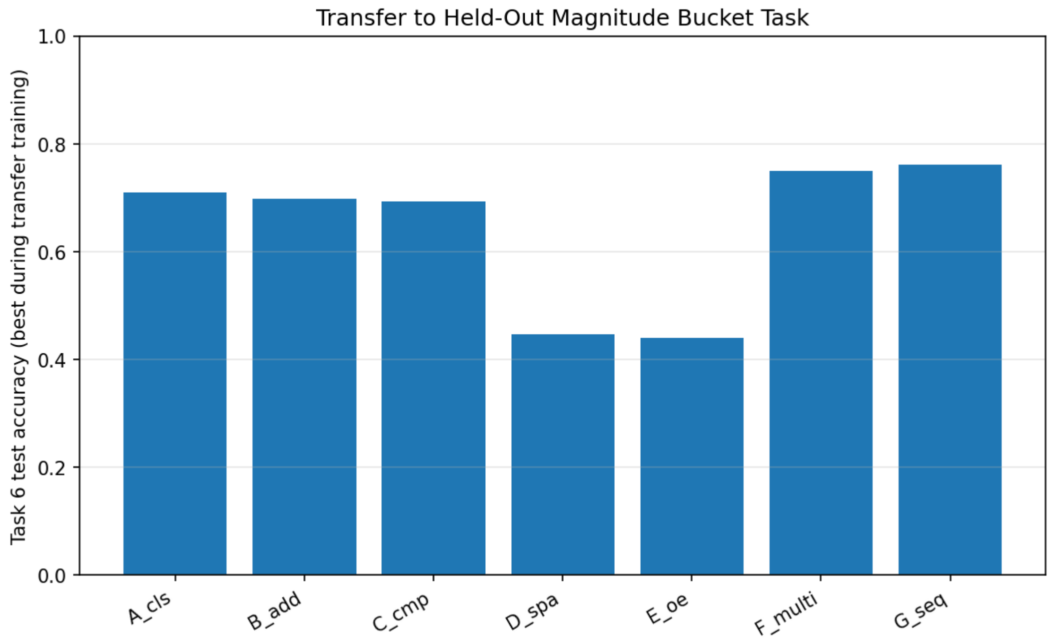

Figure 5: Transfer accuracy to the held-out magnitude bucket task across all training conditions and mixer variants.

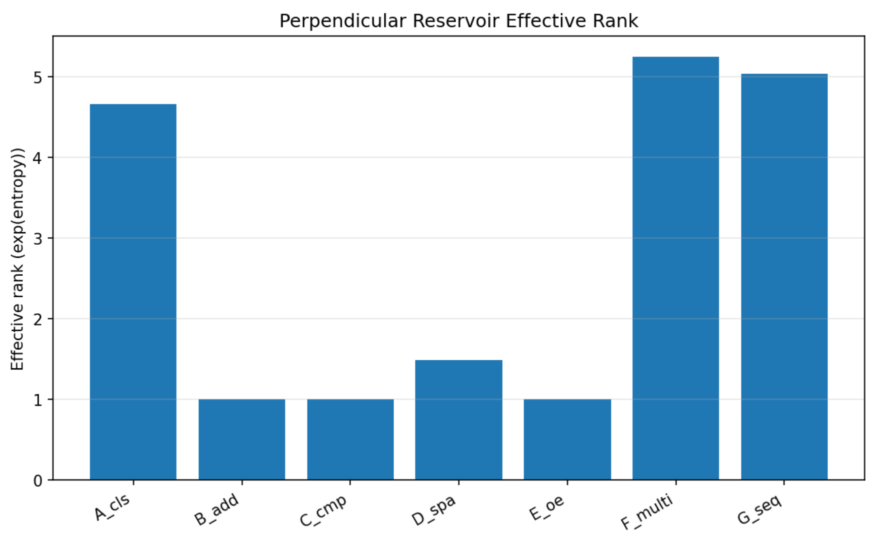

Figure 6: Pool rank across conditions. Pool rank reflects the operation's output dimensionality (scalar tasks → rank 1, classification → rank 4–6), not backbone capacity. G_seq (sequential) achieves the highest pool rank (5.45) but the worst transfer, demonstrating that pool rank does not predict transferability.

Three findings stand out:

1. Alignment, not rank, predicts transfer. All backbones hold per-half rank ∼6: the architectural capacity for digit identity is present regardless of the training task. What varies is how this capacity is *oriented*. B_add's identity manifold is aligned with digit magnitude (the transfer task needs coarse magnitude binning), yielding 0.679 transfer despite pool rank 1.0. D_spa's identity manifold is oriented toward spatial coordinates (c_x, c_y), which are nearly orthogonal to digit magnitude, hence 0.450 transfer despite comparable backbone capacity. The transfer head reads the full backbone output, bypassing the original task's pool projection entirely.

2. Coherence matters within alignment. G_seq (sequential) achieves the highest pool rank (5.45) but the worst multi-task transfer (0.681). Its high rank reflects debris from five sequential training phases, many geodesic directions populated but none maintained coherently. G_seq (cumulative) at pool rank 4.81 transfers at 0.761; F_multi at 4.74 transfers at 0.811. Among conditions with aligned identity features, the coherence of the flag structure, not the dimensionality of the pool, determines transfer quality.

3. Task-geodesic overlap dominates. The classification specialist transfers best (0.815) because magnitude bucketing (small/medium/large) is structurally closest to classification (which digit?). The specialist's identity manifold is directly aligned with the transfer task's geodesic. The multi-task conditions approach this level by accumulating multiple aligned directions, but cannot exceed the specialist when the transfer task is a near-subset of classification.

4.2 Meaning Fraction Analysis

The *meaning fraction* μ_ℓ at layer ℓ for task y quantifies the alignment between the backbone's off-diagonal energy and the task's output geodesic:

$$\mu_\ell(y) = \frac{\text{off-diagonal energy parallel to } y\text{'s geodesic at layer } \ell}{\text{total off-diagonal energy at layer } \ell} \tag{7}$$

A meaning fraction near 1.0 indicates that the layer's representation is well-aligned with the task; near 0 indicates the task's geodesic is nearly orthogonal to what the layer represents.

Condition	μ_{cls}	μ_{add}	μ_{cmp}	μ_{spa}	μ_{oe}
A: cls specialist	**0.999**	–	–	–	–
B: add specialist	–	**0.991**	–	–	–
F: Multi-task	0.998	0.915	0.977	0.138	0.999
G_seq (cumulative)	0.997	0.746	0.988	0.138	0.992
G_seq (sequential)	0.904	0.771	0.997	0.536	0.755

Table 6: Meaning fraction at layer 5 (final layer before task heads) for multi-task conditions. Single-task specialists achieve $\mu \approx 0.99$. The multi-task backbone achieves high μ for 4/5 tasks but spatial falls to $\mu = 0.138$.

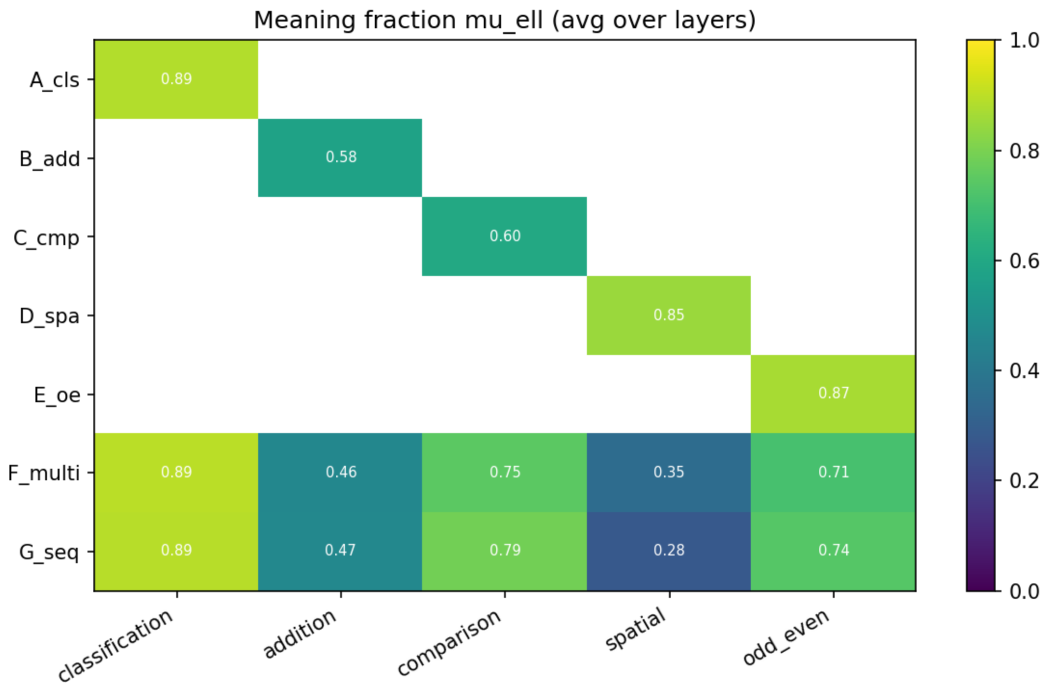

Figure 7: Meaning fraction heatmap across conditions (rows) and tasks (columns) at layer 5.

Single-task specialists converge to $\mu \approx 0.99$ for their task. Nearly all off-diagonal content is parallel to the task's output geodesic, no wasted rotation.

Spatial has $\mu = 0.138$ **in all multi-task conditions.** The spatial task's geodesic is nearly orthogonal to the shared representation at layer 5. Despite achieving near-zero MSE on-distribution, the model reads spatial information from a direction that is mostly perpendicular to the shared backbone. This predicts that spatial performance would degrade sharply under input perturbation, even though on-distribution MSE is near-zero.

Addition is partially sacrificed ($\mu = 0.746$–0.915 depending on condition). The arithmetic geodesic is not orthogonal to the shared backbone (unlike spatial) but is not fully aligned either, a consistent cost of sharing a backbone across structurally diverse tasks.

4.3 Grassmannian Occupancy

The off-diagonal energy fraction at each layer, the proportion of the representation's energy that occupies off-diagonal blocks of the Grassmannian, provides a complementary view:

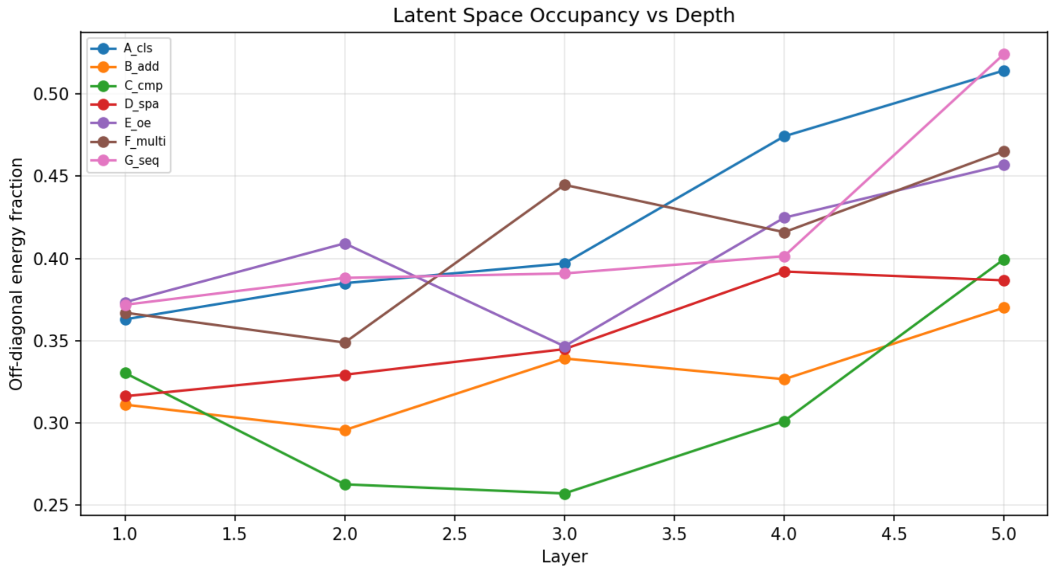

Figure 8: Off-diagonal energy fraction (Grassmannian occupancy) per layer across all conditions. Multi-task conditions converge to similar layer-5 profiles (0.469–0.483) regardless of training regime.

All multi-task conditions (F_multi, both G_seq variants) converge to the highest layer-5 occupancy (0.469–0.483), consistent with the backbone needing maximal off-diagonal content to serve multiple task heads. G_seq (cumulative) and F_multi have nearly identical layer-by-layer profiles despite different training regimes, the backbone's Grassmannian structure is determined by the task set, not the training path.

5 Cumulative Curriculum as Flag Manifold Construction

5.1 The Flag Manifold Structure

A *flag manifold* is a nested sequence of subspaces satisfying:

$$V_1 \subset V_2 \subset V_3 \subset \cdots \subset V_n, \quad \dim(V_k) = d_k < d_{k+1} \tag{8}$$

The cumulative curriculum constructs exactly this structure in the backbone's representational space:

- Phase 1: spatial only $\to$ the backbone learns V_1 (the subspace serving one task)
- Phase 2: spatial + classification $\to$ the backbone extends to $V_2 \supset V_1$
- Phase 3: + addition $\to V_3 \supset V_2$
- Phase 4: + odd/even $\to V_4 \supset V_3$
- Phase 5: + comparison $\to V_5 \supset V_4$

The nesting $V_k \subset V_{k+1}$ is maintained because all previously-active tasks remain in the loss at every phase. The optimizer cannot shrink V_k without increasing the loss on the tasks that V_k serves. Each new task can only *extend* the subspace, never contract it.

5.2 Zero Forgetting Within the Active Set

After adding	spatial	cls	add	oe	cmp
Phase 1: spatial	**0.000**	–	–	–	–
Phase 2: + cls	**0.000**	**0.670**	–	–	–
Phase 3: + add	**0.000**	**0.836**	**2.677**	–	–
Phase 4: + oe	**0.000**	**0.882**	**2.629**	**0.926**	–
Phase 5: + cmp	**0.000**	**0.891**	**2.675**	**0.938**	**0.771**

Table 7: Per-phase performance of the cumulative curriculum (mean across 3 seeds). Spatial is MSE (lower better); classification, odd/even, comparison are accuracy (higher better); addition is MAE (lower better). Dashes indicate the task head exists but was not yet in the active set. Once a task is active, its performance never decreases across subsequent phases.

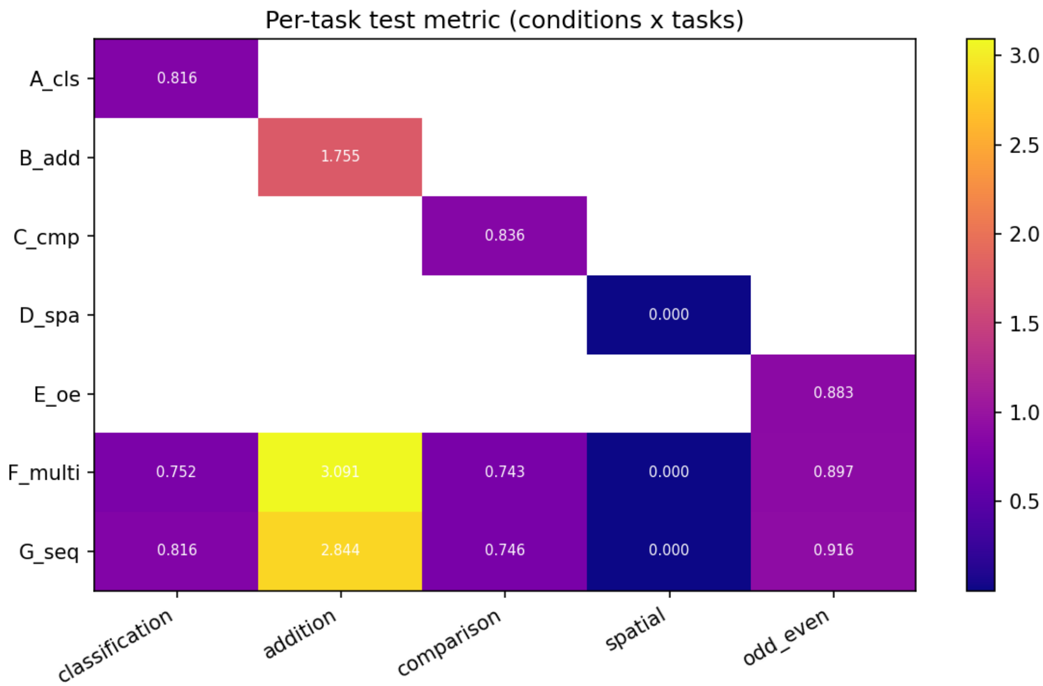

Figure 9: Per-task performance across all training conditions. Single-task specialists (left group) establish ceilings; multi-task conditions (right) show the cost and benefit of sharing a backbone.

The key observation: once a task enters the active set, its performance *never decreases* across subsequent phases. Spatial MSE holds at 0.000 through all four additional phases. Addition MAE remains at 2.63–2.68 through phases 4 and 5. This is the flag nesting in action: the subspace serving existing tasks is preserved while new dimensions are added, because the prior tasks never leave the loss.

5.3 Positive Transfer: Flag Extension Reinforces Existing Structure

Classification accuracy shows a remarkable pattern: it *improves* after its own training phase ends.

$$\text{cls accuracy:} \quad \underbrace{0.670}_{\text{end phase 2}} \rightarrow \underbrace{0.836}_{\text{after adding add}} \rightarrow \underbrace{0.882}_{\text{after adding oe}} \rightarrow \underbrace{0.891}_{\text{final}} \tag{9}$$

Each new task added to the joint objective improves the backbone's classification representations. This is the geometric signature of flag extension: adding new geodesic directions to the off-diagonal blocks populates dimensions that are partially *parallel* to classification's geodesic, not orthogonal to it. The new task's direction reinforces the existing task's alignment with the shared backbone.

This is a prediction of the Grassmannian framework that would not arise in flat-space representations. In a flat embedding space, adding new tasks can only compete with existing tasks for representational capacity. On the Grassmannian, adding new subspace directions can reinforce existing directions if their geodesics are partially aligned, extending the flag into mutually supportive dimensions.

Positive backward transfer and forgetting-free continual learning are themselves established [Lopez-Paz and Ranzato, 2017, Saha et al., 2021, Cuber and Wei, 2022], but those methods *impose* orthogonality as a constraint: gradient projection and orthogonal-subspace continual learning give a *disjoint partition* of hidden space [Chaudhry et al., 2020], not a nested flag. Here the nesting is not imposed but *measured*: keeping prior tasks in the loss suffices for the flag $V_1 \subset \cdots \subset V_5$ to appear on its own, and flags are exactly the representations a recent Grassmannian analysis argues are optimal, though it imposes them as an optimization criterion rather than observing them emerge [Szwagier and Pennec, 2025]. This measured, emergent supervised flag is the one result here we do not find already claimed.

The constructive extension also shows up across depth (Figure 10). Ranking each seed's layers by recruited capacity and normalizing to its strongest, the cumulative curriculum stays flat, its weakest layer holds 0.55 of its strongest, while simultaneous training and the specialist fall to 0.16 and 0.15, folding a layer (below 0.25 of the strongest) in 4/5 and up to four of five seeds respectively, against 1/5 for the curriculum. The evenness gap is directional at five seeds (capacity-CV $p = 0.079$): a depth-wise echo of the same extend-don't-concentrate dynamic, where building tasks in cumulatively keeps every layer engaged and concentrating them lets one fold.

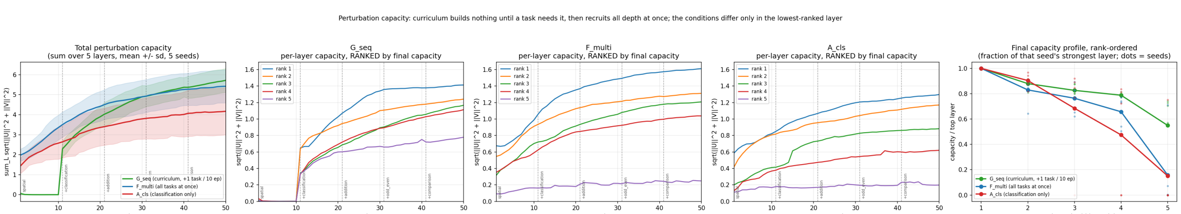

Figure 10: Recruited perturbation capacity across training (five seeds) for the classification specialist, simultaneous multi-task, and cumulative curriculum. Total recruited capacity separates the specialist from the multi-task backbones ($p = 0.016$); the rank-ordered per-layer profiles show the curriculum using its depth most evenly and the specialist folding the most layers. The folding-order and evenness gaps are directional at five seeds, not significant.

5.4 Convergence: Different Paths, Same Flag

The cumulative curriculum's final state converges to the simultaneous multi-task model on all measured axes:

Metric	F_multi	G_seq (cumulative)	Match?
cls accuracy	0.880	0.878	✓
add MAE	2.658	2.696	✓
oe accuracy	0.943	0.932	✓
Pool rank	4.74	4.81	✓
Transfer accuracy	0.811	0.761	Close
Layer-5 occupancy	0.478	0.469	✓

Table 8: Comparison of simultaneous multi-task (F_multi) and cumulative curriculum (G_seq) endpoints. The two conditions converge to structurally similar backbones despite different training paths.

The flag manifold's geometry is determined by the *task set*, not by the *order of construction*. Whether you build all five subspaces simultaneously (F_multi) or one at a time (G_seq cumulative), the endpoint is the same flag. The path through Grassmannian space differs: the curriculum traces a monotonically expanding path $\mathrm{Gr}(d_1, D) \to \mathrm{Gr}(d_2, D) \to \cdots$, while simultaneous training starts in $\mathrm{Gr}(d_5, D)$ directly, but the destination is invariant.

5.5 Sequential Replacement: Flag Destruction as Negative Control

The sequential variant (G_seq with replacement rather than accumulation) provides a clean negative control. Each phase trains on only one task, overwriting the previous:

- Classification collapses to ∼9% (random chance for 10 classes)
- Odd/even collapses to ∼50% (random chance for binary)
- Addition MAE rises to ∼11 (near-uniform guess)
- Only the last-trained task (comparison: 0.826) survives

This is flag *destruction*: each phase removes the previous task from the loss and rotates the backbone's subspace toward a new task geodesic, breaking the nesting $V_k \subset V_{k+1}$. The previous subspace V_k is not preserved; it is overwritten. The resulting representation has high pool rank (5.45) because all five training phases deposited content in the off-diagonal blocks, but the content is incoherent debris rather than a nested flag structure. Only the last phase's geodesic survives intact.

The geometric interpretation: sequential replacement traces a path through $\mathrm{Gr}(k, D)$ where each phase *jumps* to a new subspace rather than extending the existing one. The off-diagonal blocks accumulate content from all phases (hence high rank) but the meaning directions point in five different, mutually overwritten directions. No coherent flag remains.

6 Mixer Variant Comparison

6.1 The Symmetric Degradation Problem

When the antisymmetric constraint on the perturbation is removed (full_pade: $A_{\text{perturb}} = UV^T$ rather than $UV^T - VU^T$), complex tasks degrade significantly:

Task	skew_only	full_pade	pade_symreg	matexp_symreg
A: cls accuracy	**0.897**	0.816	0.889	0.880
B: add MAE	1.783	1.755	1.771	**1.758**
C: cmp accuracy	0.837	0.836	**0.839**	0.835
E: oe accuracy	**0.908**	0.883	0.907	0.906
F: multi cls	**0.880**	0.752	**0.880**	0.875

Table 9: Task performance across mixer variants (mean ± std suppressed for clarity; see Appendix). Degradation in full_pade is concentrated on complex tasks (classification, odd/even); simple tasks (addition, comparison, spatial) are unaffected.

The degradation is task-complexity dependent. Classification drops 8.1 percentage points, F_multi classification drops 12.8pp, and odd/even drops 2.5pp. Simple tasks (addition, spatial, comparison) are unaffected. The general UV^T perturbation introduces a symmetric component in A that shifts eigenvalues from the imaginary axis (rotation) toward the real axis (scaling/contraction), creating gradient instability through the coupling between cross-entropy loss and $\exp(A)$.

6.2 Symmetric Regularization Restores Performance

The pade_symreg variant adds a penalty $\lambda_{\text{sym}} \|A_{\text{sym}}\|_F^2$ where $A_{\text{sym}} = (A + A^T)/2$. This directly suppresses the symmetric eigenvalue component without constraining the skew-symmetric (rotational) structure:

- Classification: 0.816 → 0.889 (full recovery to within error bars of skew_only)
- F_multi classification: 0.752 → 0.880 (full recovery)
- F_multi transfer: 0.750 → 0.822 (exceeds skew_only's 0.811)

The symmetric penalty confirms the gradient instability hypothesis: the degradation in full_pade comes specifically from the symmetric component creating real eigenvalues, not from the perturbation direction itself. When the symmetric component is regularized to small magnitude, the additional degree of freedom (symmetric scaling alongside rotation) is mildly beneficial.

6.3 Padé Approximation Is Sufficient

The matexp_symreg variant (exact `torch.linalg.matrix_exp`) closely matches pade_symreg across all conditions. The Cayley transform (Padé [1,1] approximation to the matrix exponential) is adequate for this architecture and task set. No measurable benefit accrues from the exact computation, which costs significantly more per forward/backward pass.

6.4 Geometric Interpretation

The eigenvalue decomposition of $A = S + K$ (symmetric + skew-symmetric) reveals:

- $\exp(K)$: imaginary eigenvalues → rotation, sinusoidal structure, volume-preserving
- $\exp(S)$: real eigenvalues → magnitude scaling, contraction/expansion

The symmetric regularization penalizes S directly ($\|A_{\text{sym}}\|_F^2 = \|S\|_F^2$). With sym_reg active: the rotational axis ($\exp(K)$) is fully free to learn, while the scaling axis ($\exp(S)$) is pushed toward zero but not eliminated. This occupies a specific position in the constraint space, distinct from the L1 penalty on UV^T (which targets the magnitude of the perturbation regardless of its symmetric/skew decomposition) and from structural constraints like the convolution recovery condition (which targets the *direction* of the off-diagonal blocks).

7 Discussion

7.1 Architecture as Geometric Prior: What the Deployment Shows

Deployed in a trained multi-task system, the diagnostics support the architecture-as-geometric-prior principle without overreaching it. The same architecture, with the same initial capacity at every layer, is trained on five tasks with identical hyperparameters; the only variable is the task. Two readings are robust: tasks that are linear combinations of features (addition, spatial) need the architectural prior alone, the perturbation is driven to zero, and the audit separates genuine triviality (spatial) from proxy substitution (addition), and harder tasks recruit progressively more learned departure from the prior. The finer per-task counts, however, are seed-variable, and the reliable separation is in *total* recruited capacity rather than a sharp integer per task.

The framework predicts that such an ordering *should* exist, different tasks have different geometric complexity, and the deployment measures it, at interpretable scale, as a distribution rather than a clean staircase. Recovering an expected ordering this way is validation of the diagnostic, not the discovery of a new law.

7.2 Automatic Architecture Discovery

The L1 findings suggest a concrete approach to automatic architecture generation: start with a maximally expressive architecture (perturbation at every layer), apply L1 pressure, and read off the surviving structure. The reliable signal is coarse, whether a task set needs any learned mixer at all (for addition or spatial, none), plus the total recruited capacity, rather than an exact per-layer count, which is seed-variable; and the substitution audit must gate any "simplicity" claim. For a production system:

1. Train with L1 on perturbation parameters
2. Identify layers where perturbation is zeroed
3. Replace those layers with fixed canonical rotation (zero learned parameters)
4. Retrain the pruned architecture (optional, performance should match or improve)

For addition, this produces a pure routing + canonical rotation stack with *no learned mixer parameters*, the simplest possible architecture for the task, discovered automatically from the data's geometry.

7.3 Implications for Continual Learning

The flag manifold interpretation offers a geometric protocol for continual learning:

1. **Maintain the flag**: When adding a new task, keep all existing tasks in the loss so the optimizer preserves V_k while extending to $V_{k+1} \supset V_k$

2. **Extend, don't overwrite**: The optimizer should add new dimensions to the flag rather than rotating existing ones. The cumulative curriculum achieves this through the joint loss; structural approaches (e.g., growing the network or freezing existing parameters) could achieve it architecturally.

3. **Monitor meaning fractions**: μ_ℓ for existing tasks should not decrease when new tasks are added. A decrease signals that the new task's geodesic is being served at the expense of existing alignment.

The sequential replacement result demonstrates the cost of violating this protocol: catastrophic forgetting is not a mysterious failure mode but the predictable geometric consequence of overwriting a flag manifold's lower levels.

7.4 The Meaning Fraction as a Deployment Diagnostic

Spatial's $\mu = 0.138$ in the multi-task backbone makes a prediction: any task whose geodesic is nearly orthogonal to the backbone representation will generalize poorly under input perturbation, even when on-distribution performance appears excellent. This is a directly actionable diagnostic for deployed multi-task systems:

1. Compute μ_ℓ for each task at each layer

2. Tasks with low μ are vulnerable to distribution shift

3. Either: (a) add auxiliary tasks whose geodesics pull the backbone toward the low-μ task, or (b) add a dedicated subnetwork for the misaligned task

This diagnostic applies beyond the controlled MNIST setting to any multi-task system where tasks of different geometric character share a backbone. A low meaning fraction is a warning: the task may appear well-served on-distribution but will be fragile under perturbation or distribution shift.

7.5 Limitations

Scale. All experiments use 28×28 images, 5 tasks, state dimension 12, and ∼3,350 backbone parameters. The phenomena may not survive at transformer scale ($d = 4096$, billions of parameters). However, the quantities we measure (meaning fraction, L1-discovered complexity, flag convergence) are dimensionless ratios and structural patterns that have no inherent scale dependence.

Fixed architecture. All conditions use 5 layers of H_rotational. The theory predicts that optimal depth should vary per task (addition needs fewer layers than classification). We do not test variable-depth architectures here.

Early exit confound. The odd/even specialist early-exited at epoch 4 (accuracy ≥ 0.90 threshold). Its single-task performance (0.908) is a lower bound, not a converged value. Multi-task conditions ran for 50 full epochs. Comparisons involving odd/even should be interpreted accordingly.

Curriculum order. We test one curriculum order (spatial → cls → add → oe → cmp). The convergence result (same endpoint as F_multi) suggests order-invariance, but we have not verified this for all $5! = 120$ permutations.

Seed variance in the per-task counts. The per-task layer counts are seed-dependent. A five-seed rerun places the classification specialist at 3.6±1.1 active layers (not the 5/5 a single preliminary seed suggested) and the simultaneous multi-task condition at 4.4±0.6. What replicates is the coarse split: whether a task recruits any learned perturbation, and the total-capacity separation between conditions ($p = 0.016$). The finer orderings (e.g. fold-count differences, depth-CV at $p = 0.079$) do not reach significance at this sample size and should be read as suggestive.

L1 penalty sensitivity. The sufficiency reading is taken at one L1 weight. The zero/nonzero partition is robust near that setting, but the exact per-layer counts shift with the penalty strength; we have not mapped the full regularization path, so the counts are conditional on the reported λ.

Audit coverage. The substitution audit was run in depth only for addition (magnitude-proxy). Other zero-perturbation readings (spatial) are argued to be genuine but not audited to the same standard; a task that appears trivial may be exploiting an unexamined proxy.

8 Conclusion

We have deployed the geometric prior framework in a trained multi-task system and read three diagnostics off the weights. Each stands, once scoped to what the data support.

1. L1 regularization reads task sufficiency, and the reading must be audited. Applied to the perturbation of a canonical rotation, L1 pressure identifies which tasks the architectural prior already serves: addition and spatial center-of-mass drive the perturbation to zero at every layer. A zero is a claim to be audited, not a complexity score: it can mean genuine triviality (spatial) or proxy substitution (addition reads a magnitude axis rather than symbolic sums). The tasks that do recruit perturbation separate coarsely and reliably by whether they need a learned mixer at all; the finer per-task layer counts vary across seeds and are reported as distributions, not sharp integers.

2. Cumulative curriculum constructs a measurable flag manifold. Progressive task addition with maintenance of all prior objectives builds a nested sequence of representational subspaces ($V_1 \subset V_2 \subset \cdots \subset V_5$) where each extension reinforces existing tasks. Keeping all prior tasks in the loss preserves the nesting and yields zero forgetting; the positive backward transfer (classification improving from 0.670 to 0.891 across phases) is a known continual-learning phenomenon, here recovered as the geometric signature of flag extension populating dimensions parallel to existing geodesics. What is new is that the flag is measured off a trained backbone rather than imposed as a constraint.

3. Alignment determines transfer, not rank. All backbones hold comparable identity capacity (per-half rank $\sim$6), but transfer accuracy varies from 0.45 to 0.82 depending on whether that capacity is oriented toward the transfer task's geodesic. A spatial specialist's identity manifold is perpendicular to digit magnitude; an addition specialist's is aligned with it: same rank, very different transfer. Among aligned backbones, coherence of the flag structure further modulates quality, with incoherent debris from sequential overwriting transferring worse than a coherent flag despite higher pool rank.

These results establish measurable geometric quantities, namely meaning fraction, per-half alignment, and flag nesting, as diagnostics for multi-task representation quality. Recovering established phenomena through this lens validates the diagnostics rather than announcing a new law, and the coarse readings (does a task set need a learned mixer; is capacity aligned) are the ones that carry to continual learning and automatic architecture generation.

References

Eli Natti. Architecture as geometric prior: A primitive framework for manifold-aware neural network design. *Preprint*, 2026.

Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.

Tri Dao, Nimit Sohoni, Albert Gu, Matthew Eichhorn, Amit Blonder, Megan Leszczynski, Atri Rudra, and Christopher Ré. Kaleidoscope: An efficient, learnable representation for all structured linear maps. *ICLR*, 2020.

Mario Lezcano-Casado and David Martínez-Rubio. Cheap orthogonal constraints in neural networks: A simple parametrization of the orthogonal and unitary group. *ICML*, 2019.

Armen Aghajanyan, Sonal Gupta, and Luke Zettlemoyer. Intrinsic dimensionality explains the effectiveness of language model fine-tuning.

James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, et al. Overcoming catastrophic forgetting in neural networks. *Proc. Natl. Acad. Sci.*, 114(13):3521–3526, 2017.

David Lopez-Paz and Marc'Aurelio Ranzato. Gradient episodic memory for continual learning. *NeurIPS*, 2017.

P.-A. Absil, R. Mahony, and R. Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2008.

Alan Edelman, Tomás A. Arias, and Steven T. Smith. The geometry of algorithms with orthogonality constraints. *SIAM J. Matrix Anal. Appl.*, 20(2):303–353, 1998.

Andrew D McRae, Mark A Davenport, and Justin Romberg. Low-rank matrix recovery via iteratively reweighted least squares minimization. *SIAM J. Optim.*, 29(4):2872–2895, 2019.

Chunyuan Li, Heerad Farkhoor, Rosanne Liu, and Jason Yosinski. Measuring the intrinsic dimension of objective landscapes. *ICLR*, 2018.

Christos Louizos, Max Welling, and Diederik P. Kingma. Learning sparse neural networks through L_0 regularization. *ICLR*, 2018.

Jonathan Frankle and Michael Carbin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. *ICLR*, 2019.

Gobinda Saha, Isha Garg, and Kaushik Roy. Gradient projection memory for continual learning. *ICLR*, 2021.

Arslan Chaudhry, Naeemullah Khan, Puneet K. Dokania, and Philip H. S. Torr. Continual learning in low-rank orthogonal subspaces. *NeurIPS*, 2020.

Sen Cuber and Ying Wei. Toward sustainable continual learning: Detection and knowledge repurposing of similar tasks. *Preprint*, 2022.

Tom Szwagier and Xavier Pennec. Flag manifolds for principal subspace analysis and representation learning. *Preprint arXiv:2502*, 2025.